

HybPiper Installation Guide for Windows 11 (via WSL2)

Validated configuration

- Windows 11
- WSL 2.7.11
- Ubuntu 24.04.4 LTS
- Miniconda
- HybPiper 2.3.4

Introduction

This guide describes a validated installation procedure for HybPiper on Windows 11 using Windows Subsystem for Linux (WSL2).

1. Install WSL

Important: Start **PowerShell as Administrator** before executing the installation command.

```
winget install --id Microsoft.WSL --accept-package-agreements --accept-source-agreements
```

Verify:

```
wsl --version
```

2. Install Ubuntu

Run:

```
wsl --list --online
```

```
wsl --install -d Ubuntu-24.04
```

Important: After the Ubuntu installation has completed, **restart your computer** before continuing with the Ubuntu update and the remaining installation steps.

3. Update Ubuntu

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
sudo apt install -y wget curl git build-essential
```

4. Install Miniconda

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
```

```
bash Miniconda3-latest-Linux-x86_64.sh
```

Answer 'yes' to initialization.

```
source ~/.bashrc
```

5. Configure Conda

```
conda config --add channels bioconda
```

```
conda config --add channels conda-forge
```

```
conda config --set channel_priority strict
```

Verify that channels are ordered:

```
conda config --show channels
```

The output should be:

channels:

- conda-forge
- bioconda
- defaults

6. Install HybPiper

```
conda create -n hybpiper --override-channels -c conda-forge -c bioconda --strict-channel-priority  
python=3.12 hybpiper
```

7. Verify Installation

```
conda activate hybpiper (to activate the env)
```

```
hybpiper --version
```

```
hybpiper check_dependencies
```

Then verify:

```
bwa
```

```
samtools --version
```

```
spades.py --version
```

```
mafft --version
```

```
blastx -version
```

```
diamond version
```

```
parallel --version
```

If these outputs all show you the installed version, you are ready to start using HybPiper.

8. Overview: starting Hybpiper from scratch within your Windows OS:

Open **Windows PowerShell** or **Command Prompt** and run the beneath command to start Ubuntu:
wsl

Go to your working directory:

If this is your first session, create a dedicated working directory:

```
mkdir -p ~/hybpiper-work
```

```
cd ~/hybpiper-work
```

Important: Avoid running analyses from `/mnt/c/WINDOWS/system32` or other Windows system folders.

Activate the HybPiper environment:

```
conda activate hybpiper
```

Verify the installation:

```
hybpiper --version
```

If the version number is displayed, HybPiper is ready to use.

Troubleshooting

If 'wsl' is missing, install Microsoft.WSL using winget. Ensure WSL and VirtualMachinePlatform features are enabled. Use conda-forge before bioconda to avoid solver issues.